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FOR THE SAFETY OF AIR NAVIGATION



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REVISION SUMMARY DOCUMENT FOR THE BASE OF Aircraft DATA (BADA) REVISION 3.14

EEC Technical/Scientific Report No. 17/05/29-144

Project BADA

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TITLE:

REVISION SUMMARY DOCUMENT FOR THE BASE OF Aircraft DATA (BADA) REVISION 3.14

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Abstract:

This Revision Summary Document (RSD) describes all changes made to BADA files in Revision 3.14 since the previous release, Revision 3.13.

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SUMMARY

This Revision Summary Document (RSD) describes all changes made to BADA files in Revision 3.14 since the previous release, Revision 3.13.

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1. INTRODUCTION

1.1. IDENTIFICATION AND SCOPE

This document summarises the changes made to the Base of Aircraft Data (BADA) Revision 3.14 since the previous release, Revision 3.13.

A general description of BADA Revision 3.14 is provided in the BADA Revision 3.14 User Manual [RD1].

1.2. ORGANISATION

This document is presented in two sections including Section 1, the Introduction. This first section includes a list of referenced documents and a glossary of acronyms.

Section 2, Description of changes, presents a list of all changes to the BADA files between Revisions 3.13 and 3.14. A description for each change is also included in this section.

In addition, Appendix A presents the content of the ReleaseSummary file, which provides, for each BADA release file, information about the BADA release where this file was last modified.

1.3. REFERENCED DOCUMENTS

RD1 User Manual for the Base of Aircraft Data (BADA) Revision 3.14; EEC Technical/Scientific Report No. 17/05/29-143, June 2017.

RD2 Synonym Aircraft Report for the Base of Aircraft Data (BADA) Revision 3.14; EEC Technical/Scientific Report No. 17/05/29-147, June 2017.

RD3 Aircraft Type Designators; ICAO Document No. 8643; Edition 44, April 2016

RD4 Aircraft Type Designators; ICAO Document No. 8643; Edition 45, April 2017

1.4. GLOSSARY OF ACRONYMS

APF	**A**irline **P**rocedure **F**ile
BADA	**B**ase of **A**ircraft **Da**ta
BSA	**B**ADA **S**upport **A**pplication
CAS	**C**alibrated **Air**s**p**eed
EEC	**E**UROCONTROL **E**xperimental **C**entre
ICAO	**I**nternational **C**ivil **A**viation **O**rganisation
OPF	**O**peration **P**erformance **F**ile
PTD	**P**erformance **T**able **D**ata
PTF	**P**erformance **T**able **F**ile
RSD	**R**evision **S**ummary **D**ocument

2. DESCRIPTION OF CHANGES

Table 2-1 provides a summary of the BADA changes that were implemented between BADA Revisions 3.13 and 3.14, including the associated request identifiers in the BADA Support Application.

Table 2-1: Summary of changes for BADA Revision 3.14

Change Identifier	BSA request identifier	Title
BADA-3.14-Change-01		A10 aircraft model added
BADA-3.14-Change-02	BSA-476	A158 aircraft model added
BADA-3.14-Change-03	BSA-480	A20N aircraft model added
BADA-3.14-Change-04	BSA-288	A359 aircraft model added
BADA-3.14-Change-05		AJET aircraft model added
BADA-3.14-Change-06		C27J aircraft model added
BADA-3.14-Change-07		C310 aircraft model added
BADA-3.14-Change-08		C340 aircraft model added
BADA-3.14-Change-09		C402 aircraft model added
BADA-3.14-Change-10		C414 aircraft model added
BADA-3.14-Change-11		CL30 aircraft model added
BADA-3.14-Change-12		DA62 aircraft model added
BADA-3.14-Change-13		E45X aircraft model added
BADA-3.14-Change-14		F15 aircraft model added
BADA-3.14-Change-15		F16 aircraft model added
BADA-3.14-Change-16		F406 aircraft model added
BADA-3.14-Change-17		G150 aircraft model added
BADA-3.14-Change-18		G280 aircraft model added
BADA-3.14-Change-19		HAWK aircraft model added
BADA-3.14-Change-20	BSA-350	L39 aircraft model added
BADA-3.14-Change-21		S22T aircraft model added
BADA-3.14-Change-22	BSA-455	SVNH aircraft model added
BADA-3.14-Change-23	BSA-350	TEX2 aircraft model added
BADA-3.14-Change-24	BSA-423, 430	U2 aircraft model added
BADA-3.14-Change-25		A148 remodelling
BADA-3.14-Change-26	BSA-460	B722 remodelling
BADA-3.14-Change-27		C421 remodelling
BADA-3.14-Change-28		C750 remodelling

BADA-3.14-Change-29		E120 remodelling
BADA-3.14-Change-30		MU2 remodelling
BADA-3.14-Change-31		P180 remodelling
BADA-3.14-Change-32	BSA-228	SW4 remodelling
BADA-3.14-Change-33		E170 split into E170, E75S and E75L
BADA-3.14-Change-34	BSA-471	E190 split into E190 and E195
BADA-3.14-Change-35		FGTN model removed
BADA-3.14-Change-36		B748 maximum altitude increase
BADA-3.14-Change-37	BSA-421	DH8D maximum altitude increase
BADA-3.14-Change-38	BSA-350, 407, 408, 409, 412, 413, 415, 420, 423, 424, 442, 443, 470, 482, 484	Revision of synonym aircraft types
BADA-3.14-Change-39	BSA-406, 473, 475	Documentation update

The details of each change are presented in the subsequent pages:

- For each change, a description of the change, a motivation for the change and a specification of the affected files are provided. For an explanation of symbols related to BADA coefficients, refer to [RD1].
- If either the OPF or APF file has been changed then the PTF and PTD files are re-created with the updated files.
- For aircraft models whose update impacts vertical performances, a plot is provided to give an overview of the impact of the changes on typical climb and descent profiles. This plot has been generated using the following assumptions:
    - Three climb profiles are provided, corresponding to three values of the aircraft mass: low (blue line), nominal (green line) and high (red line);
    - One descent profile is provided, corresponding to the nominal aircraft mass (light blue);
    - Climb and descent trajectories are flown according to the Airline Procedure Model described in [RD1];
    - The behaviour of the previous version of the model is presented in dashed lines, while the behaviour of the new model is presented in full lines.

BADA-3.14-CHANGE-01

A10 aircraft model added

Description of Modification:

A new BADA model for the Fairchild A-10A aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

A10___.APF	added
A10___.OPF	added
A10___.PTF	added
A10___.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-02

A158 aircraft model added

Description of Modification:

A new BADA model for the Antonov An-158 aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

A158__.APF	added
A158__.OPF	added
A158__.PTF	added
A158__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-03

A20N aircraft model added

Description of Modification:

A new BADA model for the Airbus A320-271N “Neo” aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

A20N__.APF	added
A20N__.OPF	added
A20N__.PTF	added
A20N__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-04

A359 aircraft model added

Description of Modification:

A new BADA model for the Airbus A350-941 aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

A359__.APF	added
A359__.OPF	added
A359__.PTF	added
A359__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-05

AJET aircraft model added

Description of Modification:

A new BADA model for the Dassault-Dornier Alpha Jet aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

AJET__.APF	added
AJET__.OPF	added
AJET__.PTF	added
AJET__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-06

C27J aircraft model added

Description of Modification:

A new BADA model for the Alenia C-27J aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

C27J__.APF	added
C27J__.OPF	added
C27J__.PTF	added
C27J__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-07

C310 aircraft model added

Description of Modification:

A new BADA model for the Cessna 310R aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

C310__.APF	added
C310__.OPF	added
C310__.PTF	added
C310__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-08

C340 aircraft model added

Description of Modification:

A new BADA model for the Cessna 340A aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

C340__.APF	added
C340__.OPF	added
C340__.PTF	added
C340__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-09

C402 aircraft model added

Description of Modification:

A new BADA model for the Cessna 402B aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

C402__.APF	added
C402__.OPF	added
C402__.PTF	added
C402__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-10

C414 aircraft model added

Description of Modification:

A new BADA model for the Cessna 414A aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

C414__.APF	added
C414__.OPF	added
C414__.PTF	added
C414__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-11

CL30 aircraft model added

Description of Modification:

A new BADA model for the Bombardier Challenger 300 has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

CL30__.APF	added
CL30__.OPF	added
CL30__.PTF	added
CL30__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-12

DA62 aircraft model added

Description of Modification:

A new BADA model for the Diamond DA62 aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

DA62__.APF	added
DA62__.OPF	added
DA62__.PTF	added
DA62__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-13

E45X aircraft model added

Description of Modification:

A new BADA model for the Embraer EMB-145XR aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

E45X__.APF	added
E45X__.OPF	added
E45X__.PTF	added
E45X__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-14

F15 aircraft model added

Description of Modification:

A new BADA model for the McDonnell-Douglas F-15E aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

F15___.APF	added
F15___.OPF	added
F15___.PTF	added
F15___.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-15

F16 aircraft model added

Description of Modification:

A new BADA model for the General Dynamics F-16C aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

F16___.APF	added
F16___.OPF	added
F16___.PTF	added
F16___.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-16

F406 aircraft model added

Description of Modification:

A new BADA model for the Cessna 406 aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

F406__.APF	added
F406__.OPF	added
F406__.PTF	added
F406__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-17

G150 aircraft model added

Description of Modification:

A new BADA model for the Gulfstream G150 aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

G150__.APF	added
G150__.OPF	added
G150__.PTF	added
G150__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-18

G280 aircraft model added

Description of Modification:

A new BADA model for the Gulfstream G280 aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

G280__.APF	added
G280__.OPF	added
G280__.PTF	added
G280__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-19

HAWK aircraft model added

Description of Modification:

A new BADA model for the British Aerospace T-45A Goshawk aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

HAWK__.APF	added
HAWK__.OPF	added
HAWK__.PTF	added
HAWK__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-20

L39 aircraft model added

Description of Modification:

A new BADA model for the Aero L-39 Albatros aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

L39___.APF	added
L39___.OPF	added
L39___.PTF	added
L39___.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-21

S22T aircraft model added

Description of Modification:

A new BADA model for the Cirrus SR22T aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

S22T__.APF	added
S22T__.OPF	added
S22T__.PTF	added
S22T__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-22

SVNH aircraft model added

Description of Modification:

A new BADA model for the ICP Savannah VG aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

SVNH__.APF	added
SVNH__.OPF	added
SVNH__.PTF	added
SVNH__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-23

TEX2 aircraft model added

Description of Modification:

A new BADA model for the Raytheon T-6B Texan II aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

TEX2__.APF	added
TEX2__.OPF	added
TEX2__.PTF	added
TEX2__.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-24

U2 aircraft model added

Description of Modification:

A new BADA model for the Lockheed U-2 aircraft type has been added.

Reason for Modification:

Following the availability of aircraft performance reference data, the decision has been taken to model this aircraft type.

Files Affected:

U2____.APF	added
U2____.OPF	added
U2____.PTF	added
U2____.PTD	added
SYNONYM.NEW	modified

BADA-3.14-CHANGE-25

A148 remodelling

Description of Modification:

The BADA model for the Antonov An-148 aircraft type has been redeveloped.

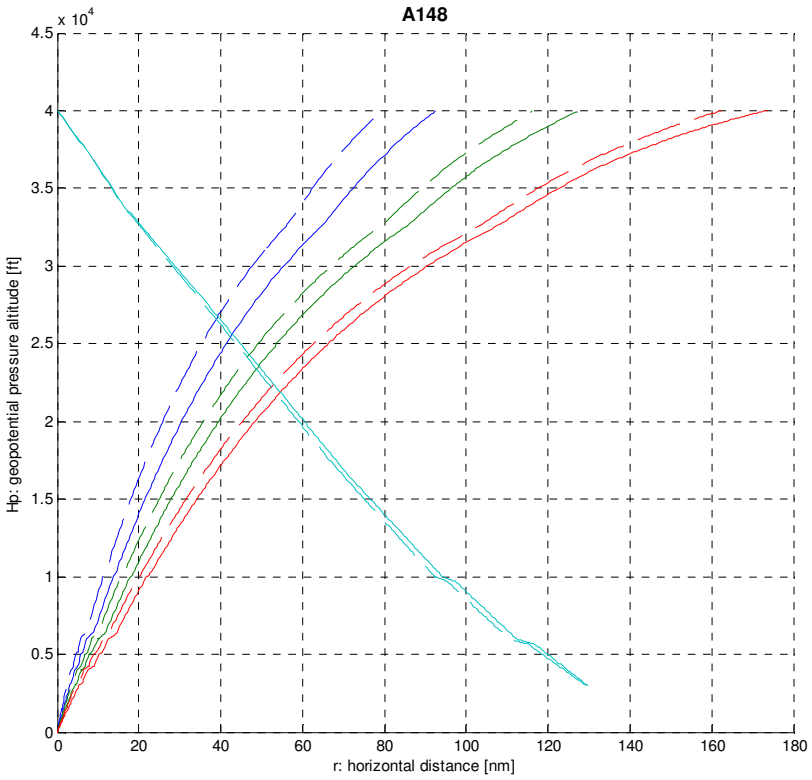
Reason for Modification:

Following the availability of updated aircraft performance reference data for this aircraft, the decision has been taken to remodel this aircraft type.

Files Affected:

A148__.APF	modified
A148__.OPF	modified
A148__.PTF	modified
A148__.PTD	modified

Impact on Flight Profiles:



BADA-3.14-CHANGE-26

B722 remodelling

Description of Modification:

The BADA model for the Boeing 727-228 aircraft type has been redeveloped. In particular, the maximum operating altitude has been increased from 37,000ft to 42,000ft.

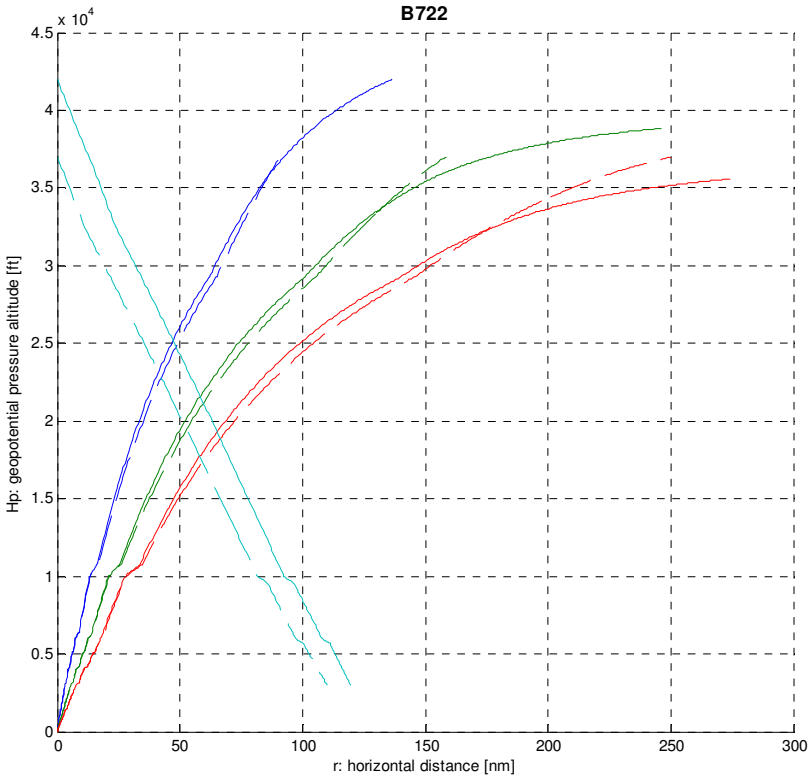
Reason for Modification:

Following the availability of updated aircraft performance reference data for this aircraft, the decision has been taken to remodel this aircraft type.

Files Affected:

B722__.APF	modified
B722__.OPF	modified
B722__.PTF	modified
B722__.PTD	modified

Impact on Flight Profiles:



BADA-3.14-CHANGE-27

C421 remodelling

Description of Modification:

The BADA model for the Cessna 421 aircraft type has been redeveloped. The new model is based on the 421C version, whereas the old one was based on the 421B version.

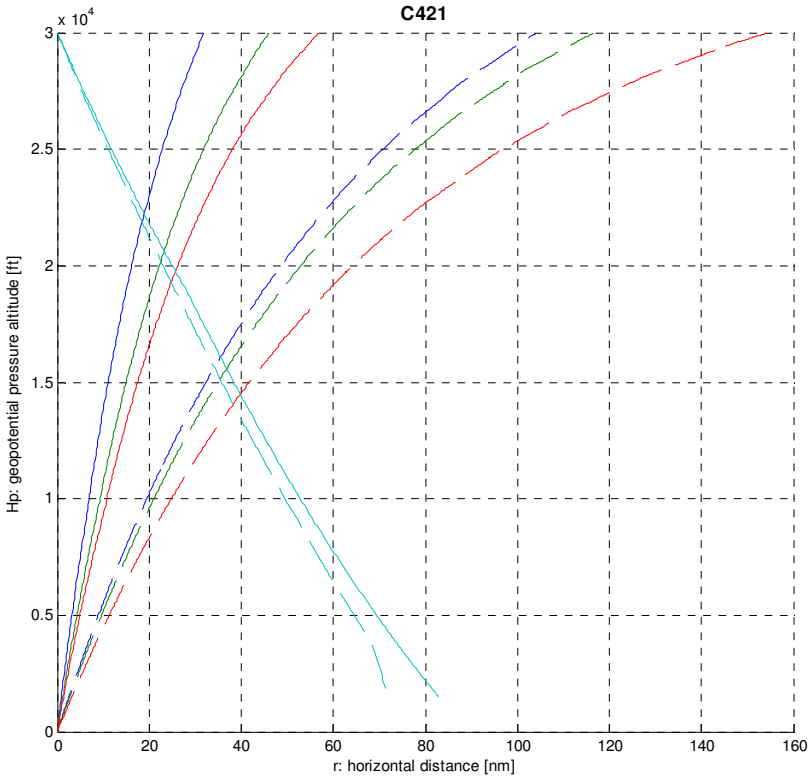
Reason for Modification:

Following the availability of updated aircraft performance reference data for this aircraft, the decision has been taken to remodel this aircraft type.

Files Affected:

C421__.APF	modified
C421__.OPF	modified
C421__.PTF	modified
C421__.PTD	modified

Impact on Flight Profiles:



BADA-3.14-CHANGE-28

C750 remodelling

Description of Modification:

The BADA model for the Cessna Citation X (Model 750) aircraft type has been redeveloped.

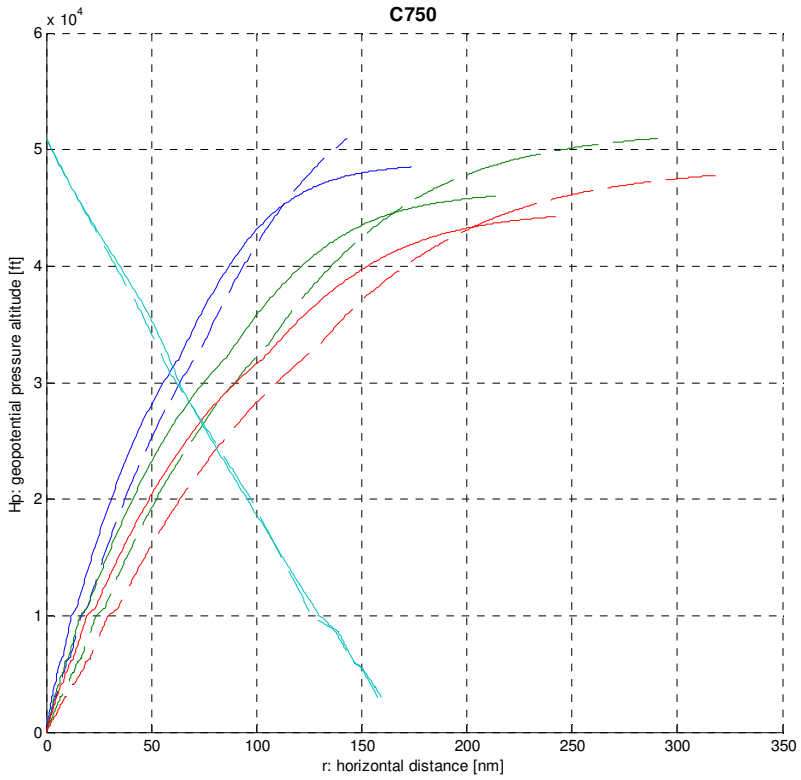
Reason for Modification:

Following the availability of updated aircraft performance reference data for this aircraft, the decision has been taken to remodel this aircraft type.

Files Affected:

C750__.APF	modified
C750__.OPF	modified
C750__.PTF	modified
C750__.PTD	modified

Impact on Flight Profiles:



BADA-3.14-CHANGE-29

E120 remodelling

Description of Modification:

The BADA model for the Embraer EMB-120 aircraft type has been redeveloped.

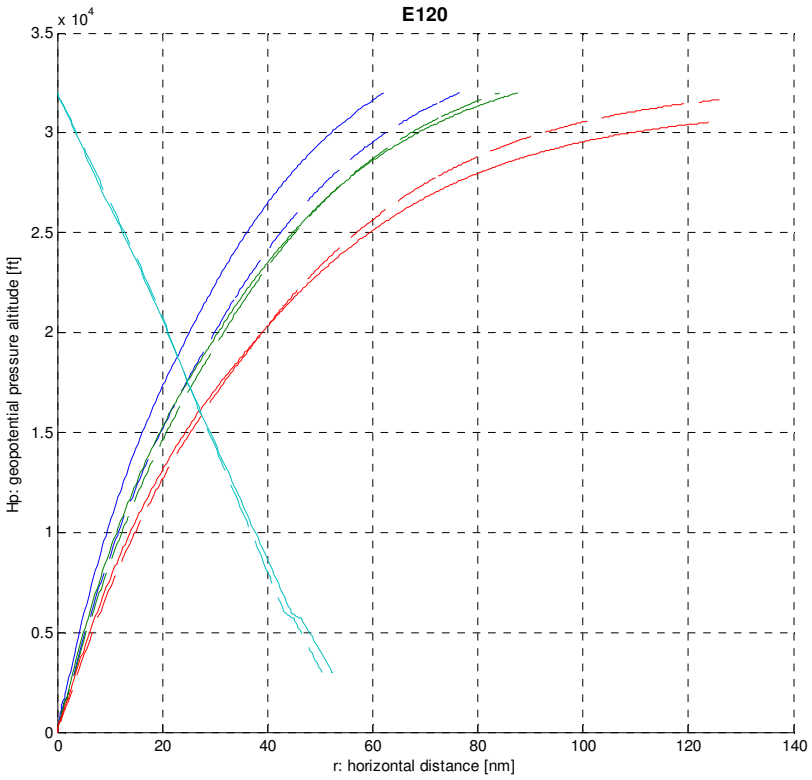
Reason for Modification:

Following the availability of updated aircraft performance reference data for this aircraft, the decision has been taken to remodel this aircraft type.

Files Affected:

E120__.APF	modified
E120__.OPF	modified
E120__.PTF	modified
E120__.PTD	modified

Impact on Flight Profiles:



BADA-3.14-CHANGE-30

MU2 remodelling

Description of Modification:

The BADA model for the Mitsubishi MU-2 aircraft type has been redeveloped. The new model is based on the MU-2B-60 version, whereas the old one was based on the MU-2B-25A version. In particular, the maximum operating altitude has been increased from 28,000ft to 31,000ft.

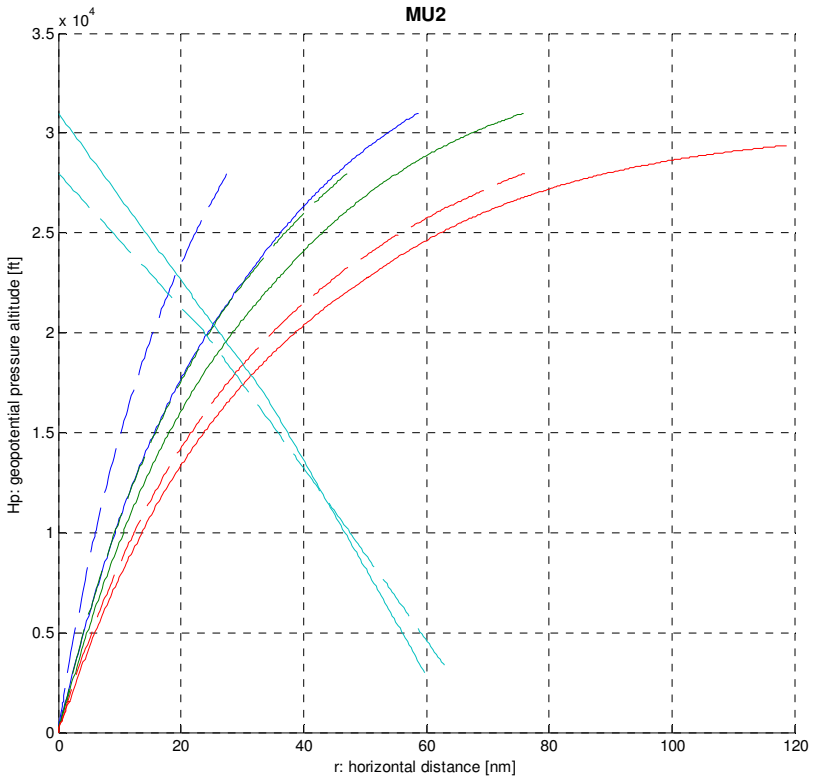
Reason for Modification:

Following the availability of updated aircraft performance reference data for this aircraft, the decision has been taken to remodel this aircraft type.

Files Affected:

MU2___.APF	modified
MU2___.OPF	modified
MU2___.PTF	modified
MU2___.PTD	modified

Impact on Flight Profiles:



BADA-3.14-CHANGE-31

P180 remodelling

Description of Modification:

The BADA model for the Piaggio P.180 aircraft type has been redeveloped. The new model is based on the Avanti II version, whereas the old one was based on the Avanti version.

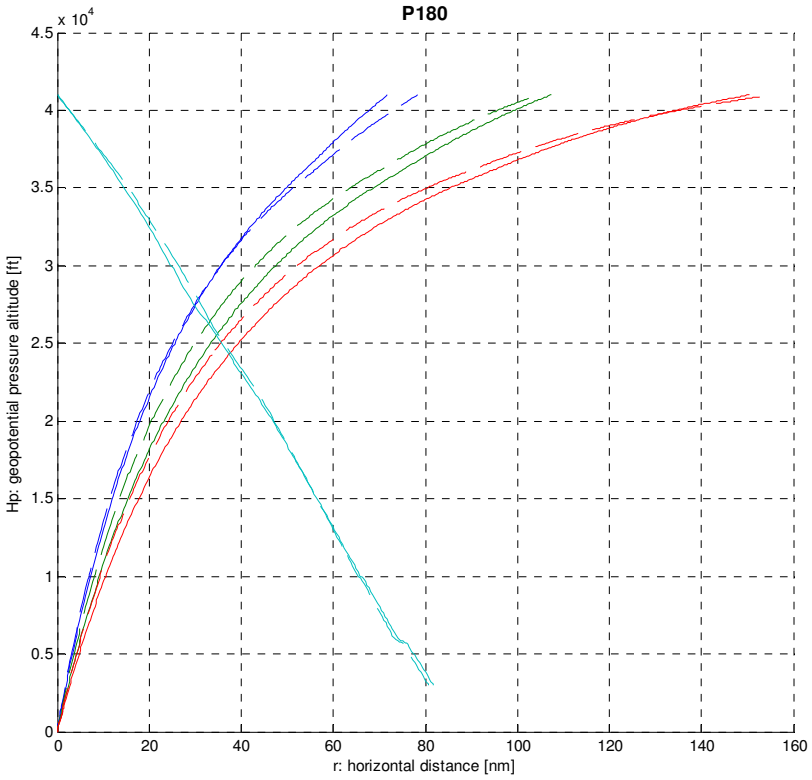
Reason for Modification:

Following the availability of updated aircraft performance reference data for this aircraft, the decision has been taken to remodel this aircraft type.

Files Affected:

P180__.APF	modified
P180__.OPF	modified
P180__.PTF	modified
P180__.PTD	modified

Impact on Flight Profiles:



BADA-3.14-CHANGE-32

SW4 remodelling

Description of Modification:

The BADA model for the Fairchild/Swearingen Metro aircraft type has been redeveloped. The new model is based on the SA227-AC version, whereas the old one was based on the SA226-TC version. In particular, the maximum operating altitude has been increased from 25,000ft to 31,000ft.

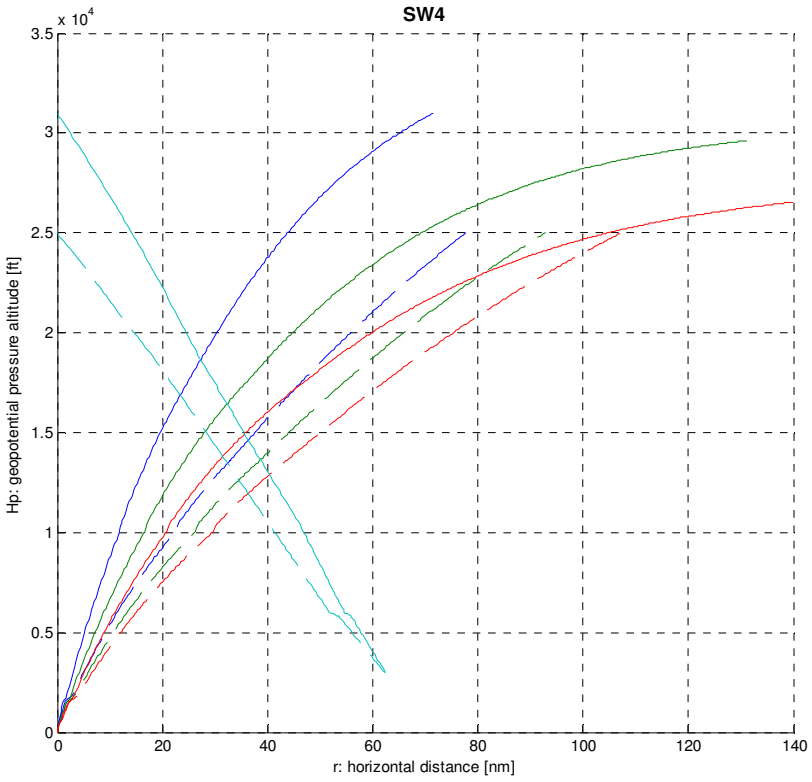
Reason for Modification:

Following the availability of updated aircraft performance reference data for this aircraft, the decision has been taken to remodel this aircraft type.

Files Affected:

SW4___.APF	modified
SW4___.OPF	modified
SW4___.PTF	modified
SW4___.PTD	modified

Impact on Flight Profiles:



BADA-3.14-CHANGE-33

E170 split into E170, E75S and E75L

Description of Modification:

1. A new BADA model for the Embraer 175 (short wing) aircraft type has been added.
2. A new BADA model for the Embraer 175 (long wing) aircraft type has been added.
3. The BADA model for the Embraer 170 aircraft type has been updated.

Reason for Modification:

Following the split of the E170 designator into E170, E75S and E75L, introduced by [RD3], the Embraer 175 with short wings (resp. with long wings) is now assigned the E75S (resp. E75L) designator while the Embraer 170 retains the E170 designator. Together with the development of new models for the similar Embraer 175 aircraft types, the decision has been taken to remodel the Embraer 170 aircraft type to ensure as good a consistency as possible between these models.

Files Affected:

E75S__.APF	added
E75S__.OPF	added
E75S__.PTF	added
E75S__.PTD	added
E75L__.APF	added
E75L__.OPF	added
E75L__.PTF	added
E75L__.PTD	added
E170__.APF	modified
E170__.OPF	modified
E170__.PTF	modified
E170__.PTD	modified
SYNONYM.NEW	modified

BADA-3.14-CHANGE-34

E190 split into E190 and E195

Description of Modification:

1. A new BADA model for the Embraer 195 aircraft type has been added.
2. The BADA model for the Embraer 190 aircraft type has been updated.

Reason for Modification:

Following the split of the E190 designator into E190 and E195, introduced by [RD4], the Embraer 195 is now assigned the E195 designator while the Embraer 190 retains the E190 designator. Together with the development of a new model for the similar Embraer 195 aircraft type, the decision has been taken to remodel the Embraer 190 aircraft type to ensure as good a consistency as possible between these models.

Files Affected:

E195__.APF	added
E195__.OPF	added
E195__.PTF	added
E195__.PTD	added
E190__.APF	modified
E190__.OPF	modified
E190__.PTF	modified
E190__.PTD	modified
SYNONYM.NEW	modified

BADA-3.14-CHANGE-35

FGTN model removed

Description of Modification:

The BADA model for the generic “normal performances” fighter aircraft has been removed.

Reason for Modification:

Following the addition of BADA models for several fighter aircraft types, covering a wide range of aircraft sizes and “normal” performances (representative of general air traffic operations), the decision has been taken to remove this generic model. The other generic fighter models, FGTH (high performances, e.g. using afterburner, representative of specific fighter operations) and FGTL (low performances, representative of long range/bomber operations), are maintained as they represent different types of operations.

Files Affected:

FGTN__.APF	removed
FGTN__.OPF	removed
FGTN__.PTF	removed
FGTN__.PTD	removed
SYNONYM.NEW	modified

BADA-3.14-CHANGE-36

B748 maximum altitude increase

Description of Modification:

The maximum operating altitude has been increased from 42,100ft to 43,100ft in the BADA model for the Boeing 747-8(F) aircraft type.

Reason for Modification:

While the freighter version (747-8F), on which the B748 model was originally based, is certified up to a maximum operating altitude of 42,100ft, the passenger version (747-8) is certified up to a maximum operating altitude of 43,100ft. Since both versions share the same ICAO designator, it has been decided to increase the maximum operating altitude of the B748 model to the highest of those two values.

Files Affected:

B748__.OPF	modified
B748__.PTF	modified
B748__.PTD	modified

BADA-3.14-CHANGE-37

DH8D maximum altitude increase

Description of Modification:

The maximum operating altitude has been increased from 25,000ft to 27,000ft in the BADA model for the Dash 8 Q400 aircraft type.

Reason for Modification:

A high-altitude version of the Dash 8 Q400 with drop-down oxygen masks is available, which increases maximum operating altitude to 27,000ft.

Files Affected:

DH8D__.OPF	modified
DH8D__.PTF	modified
DH8D__.PTD	modified

BADA-3.14-CHANGE-38

Revision of synonym aircraft types

Description of Modification:

The list of synonyms has been updated to take into account the following changes:

1) Addition of all aircraft types present in the European traffic from 2000 to 2016, and most important aircraft types present in the USA and Australian traffic, which were not available as synonyms.
2) Application of ICAO designator changes (additions and deletions) introduced by Editions 44 [RD3] and 45 [RD4] of ICAO Doc 8643.
3) Addition of user-requested synonyms.
4) Review and correction of existing data in the NAVION database used as input to map the synonyms to BADA native models.
5) Inclusion of the wingspan parameter in the weighting function used to map the synonyms.

Those changes result in 567 additions, 112 modifications and 1 deletion of synonyms. More details can be found in [RD2].

Reason for Modification:

This revision is twofold: it allows for the addition of new synonym aircraft, and also a general reevaluation of all previously assigned aircraft to account for the addition of the new BADA models in the new release.

Files Affected:

SYNONYM.NEW	modified

BADA-3.14-CHANGE-39

Documentation update

Description of Modification:

The following changes have been made to the aircraft performance model algorithms described in the User Manual [RD1]:

1. Correction of the formulas used to compute the CAS/Mach transition altitude (Section 3.1.2.g).
2. Extension of the low speed buffeting limit and associated coefficients to turboprop aircraft (Sections 3.6.2 and 3.11).

Those changes are susceptible to require modifications in BADA software implementations to ensure the conformance of such implementations to the aircraft performance model of BADA Revision 3.14.

Reason for Modification:

1. The previous formulas did not provide the correct results when the transition altitude was located above the tropopause: the computation of the pressure ratio at the transition altitude (formula 3.1-29) was correct, but the computation of the corresponding pressure altitude was not (formulas 3.1-27/28).
2. At least one turboprop aircraft (P180) has been identified as subject to buffeting, as indicated by the buffet boundary data presented in its flight manual. Buffet coefficients were active only for jet aircraft in the BADA 3.13 specifications: this behaviour had to be extended to turboprop aircraft to take into account the case of the P180, and possibly other aircraft in the future.

Files Affected:

BADA User Manual	modified.

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APPENDIX A – RELEASE SUMMARY FILE

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BADA Release Summary File
Summary Date:  Jun 29 2017

   BADA Release:  3.14


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File Name        ReleaseID
==========================
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A10___.PTD       3.14
A10___.PTF       3.14
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F28___.PTF	3.8
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F2TH__.PTF	3.9
F406__.APF	3.14
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F406__.PTF	3.14
F50___.APF	3.10
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F50___.PTF	3.10
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F70___.PTF	3.11
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F900__.PTF	3.9
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FA10__.PTF	3.8
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FA20__.PTF	3.9
FA50__.APF	3.8
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FA50__.PTF	3.9
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MAGC__.PTF	3.13
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MD11__.PTF	3.8
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MU2___.PTF	3.14
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P180__.PTF	3.14
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P28U__.PTF	3.12
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P46T__.PTF	3.12

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PA27__.PTF	3.9
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PA44__.PTF	3.11
PA46__.APF	3.11
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PA46__.PTF	3.11
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PAY2__.PTF	3.8
PAY3__.APF	3.7
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PAY3__.PTF	3.8
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PC12__.PTF	3.12
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S22T__.PTF	3.14
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SF25__.PTF	3.13
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SH36__.PTD	3.8
SH36__.PTF	3.8
SR20__.APF	3.13

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SR22__.PTF      3.10
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SU95__.PTF      3.12
SVNH__.APF      3.14
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SVNH__.PTF      3.14
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SW4___.PTF      3.14
SYNONYM.NEW     3.14
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T134__.PTF      3.8
T154__.APF      3.7
T154__.OPF      3.7
T154__.PTD      3.8
T154__.PTF      3.8
T204__.APF      3.11
T204__.OPF      3.11
T204__.PTD      3.11
T204__.PTF      3.11
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TB20__.OPF      3.12
TB20__.PTD      3.12
TB20__.PTF      3.12
TB21__.APF      3.12
TB21__.OPF      3.12
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TB21__.PTF      3.12
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TEX2__.PTF      3.14
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U2____.PTF      3.14
YK40__.APF      3.11
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YK40__.PTF      3.12
YK42__.APF      3.11
YK42__.OPF      3.11
YK42__.PTD      3.11
YK42__.PTF      3.11
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